

8(a)(i)	$v_N = 3.4 \times 10^7 \times \sin 30^\circ$ $= 1.7 \times 10^7 \text{ m s}^{-1}$	A1
8(a)(ii)	$mv^2 / r = Bqv$ or $r = mv / Bq$	C1
	$r = (9.11 \times 10^{-31} \times 1.7 \times 10^7) / (3.2 \times 10^{-3} \times 1.60 \times 10^{-19})$	C1
	$= 0.030 \text{ m}$	A1
8(b)	zero	B1
8(c)	helix/coil	B1

Question	Answer	Marks
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